



THREE-PHASE
HYBRID INVERTER

HI-80-125K-TH

-  IP66 rated enclosure, designed for both indoor and outdoor deployment in harsh environments.
-  Supports 100% 3-phase unbalanced output on both ports (BACKUP & GRID).
-  Integrated 0-export function.
-  Sustain 200% rated power overload for 10 seconds.
-  Peak shaving and valley filling to enhance grid stability.
-  110% continuous overload rating at sustained operation.

Technical Data	HI-80K-TH	HI-90K-TH	HI-100K-TH	HI-110K-TH	HI-125K-TH
INPUT DC (PV SIDE)					
Max. PV access power	160kW	180kW	200kW	220kW	250kW
Max. PV input power	128kW	144kW	160kW	176kW	200kW
Max. PV input voltage	1000V				
Nominal input voltage	600V				
MPPT working voltage range	180V-850V				
Number of independent MPPT / strings per MPPT	8/3				
MPPT max. current	8×45A				
Start-up voltage	200V				
OUTPUT/INPUT AC (GRID SIDE)					
Nominal apparent power	80kW	90kW	100kW	110kW	125kW
Max. apparent power	88kVA	100kVA	110kVA	120kVA	137kVA
Nominal voltage	380V/400V				
Nominal frequency	50Hz/60Hz				
Rated current	121A/116A	136A/130A	151A/145A	167A/159A	189A/181A
Max. current	133A	149A	166A	184A	207A
Power factor	-0.8~+0.8 (adjustable)				
THDi	<3% (Nominal Output)				
Grid system pattern	3L+N+PE				
Max. AC current from utility grid	300A				
OUTPUT AC (BACK-UP)					
Nominal output power	80kW	90kW	100kW	110kW	125kW
Max. output apparent power	88kVA	100kVA	110kVA	120kVA	137kVA
Nominal output voltage	380V/400V				
Nominal output frequency	50Hz/60Hz				
Output THDV	<3%				
Automatic switch time	<10ms				
BATTERY					
Battery type	Li-ion				
Battery voltage range	200V-800V				
Max. charging/discharging current	2×100A / 2×100A				
Charging strategy for Li-ion battery	Self-adaption to BMS				
EFFICIENCY					
Max. MPPT efficiency	99.9%				
Europe efficiency	97.5%				
Max. battery to load efficiency	98%				
PROTECTION					
Anti-islanding protection	Yes				
Insulation impedance protection	Yes				
RCD detection	Yes				
PV reverse polarity protection	Yes				
Output over voltage protection	Yes				
Output over current protection	Yes				
GENERAL DATA					
Dimensions(L/W/H) in mm	1140/367/820				
Weight	137kg				
Operation temperature range	-30℃~+60℃				
Heat dissipation mode	Smart cooling				
IP Class	IP 66				
Maximum altitude	4000m				
Self-Consumption night	<20W				
Topology	Transformerless				
FEATURES					
LCD display	Yes				
Communication interface	WiFi/GPRS/RS485				